

ARTCONNECTPRO-APP PROJECT

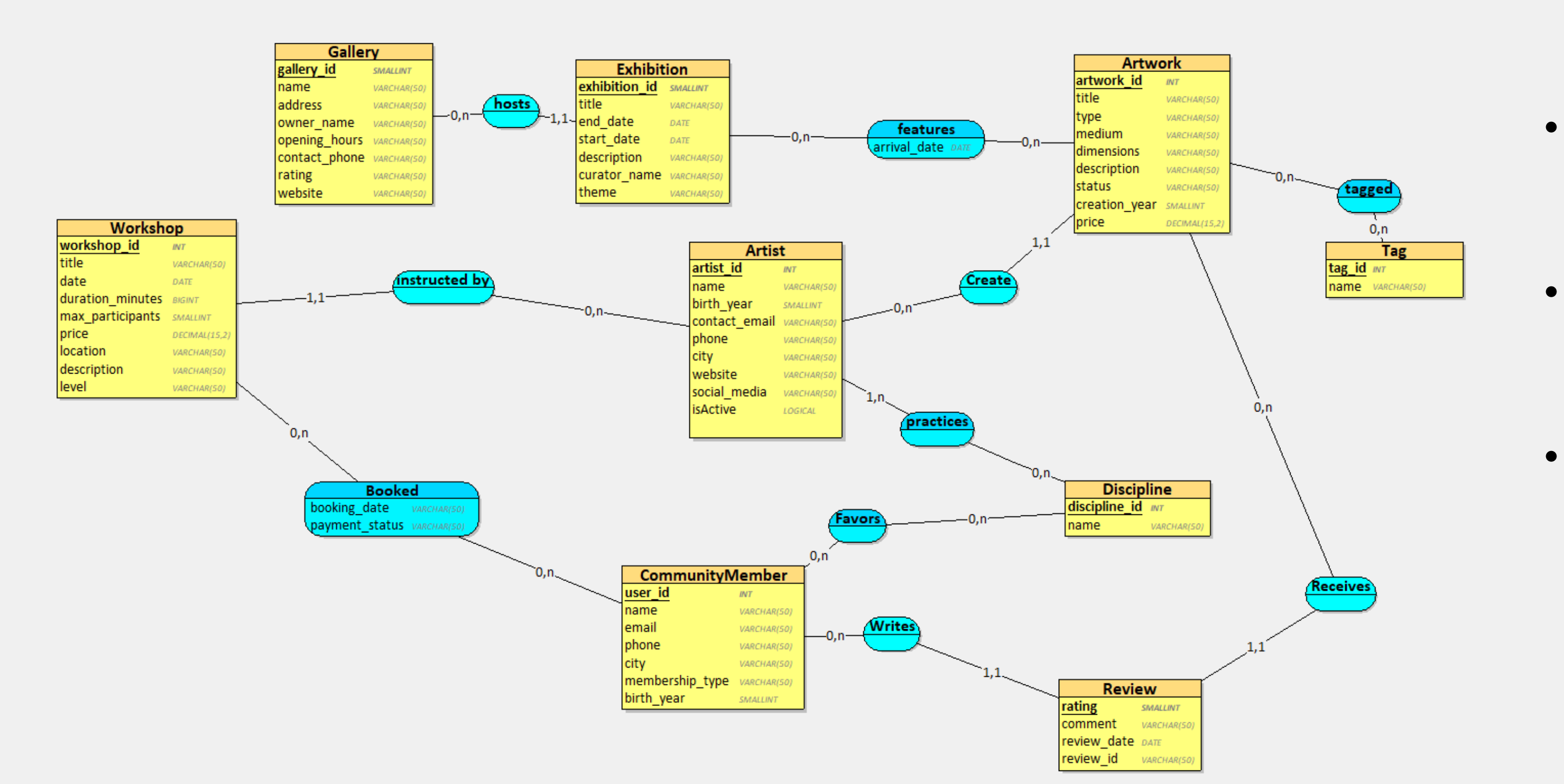
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Promotion 2028

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Database Design & Normalization

What we normalized:



- No duplication of core data
Artist info only in Artist,
gallery info only in Gallery
- All many-to-many resolved,
tagged, practices, favors,
features
- No derived data stored
Participation counts
computed via
views/functions

The SQL Engine (Advanced Features)

Views

view_artwork_catalog_with_artist

```
String sql = "SELECT a.*, art.name AS  
artist_name FROM Artwork a JOIN Artist art  
ON a.artist_id = art.artist_id WHERE a.title = ?";
```

```
String sql = "SELECT * FROM  
view_artwork_catalog_with_artist WHERE title  
= ?";
```

Procedure / Trigger

create_workshop_with_artist

- Prevents creation of workshop be created without an instructor

book_multiple_workshops_transaction

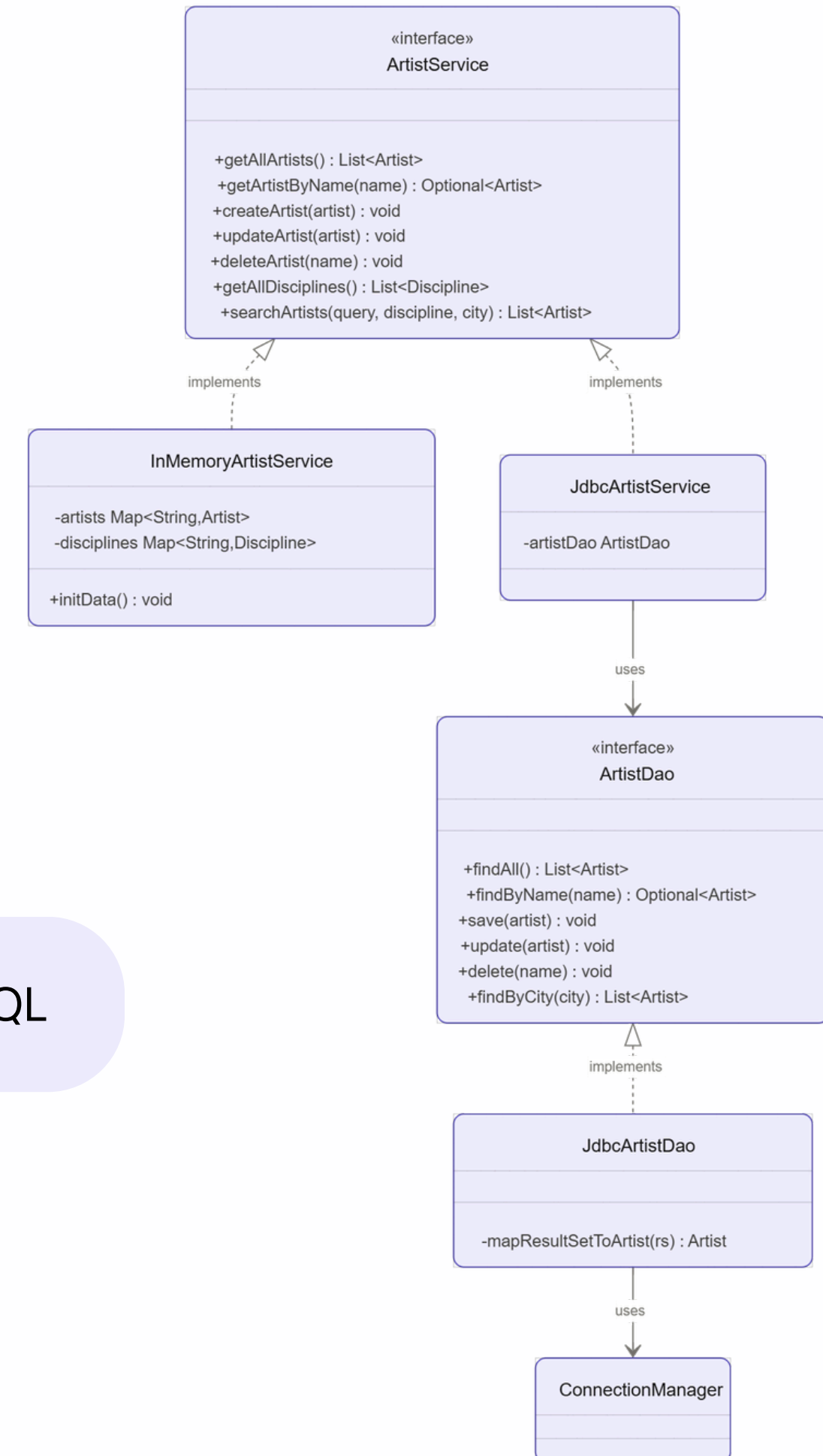
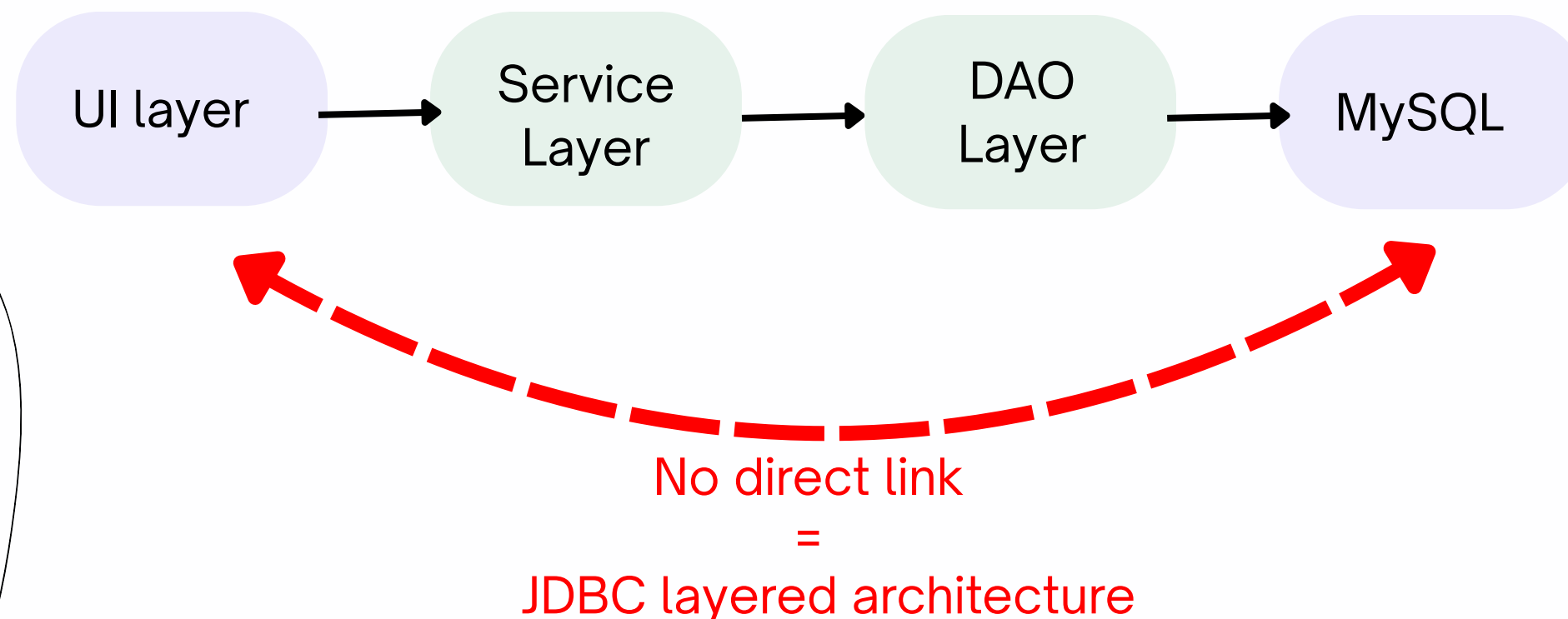
- If a user tries to book three different classes at once, this transaction guarantees they get a spot in all three, or the entire checkout cancels

The java architecture

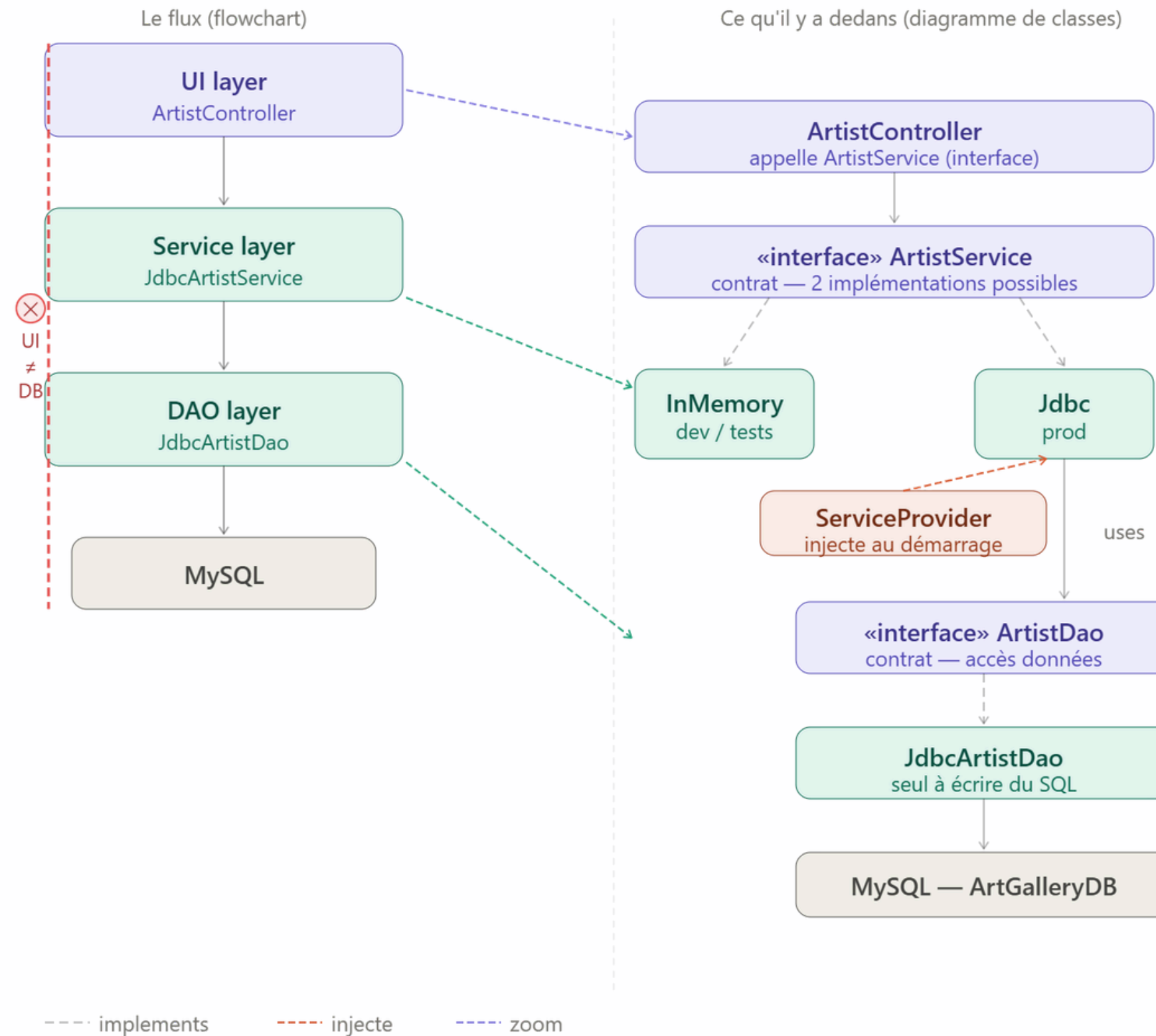
Correct required layered architecture :

- UI layer never talks to the database directly ;
- Every call flows through the following flow chart.

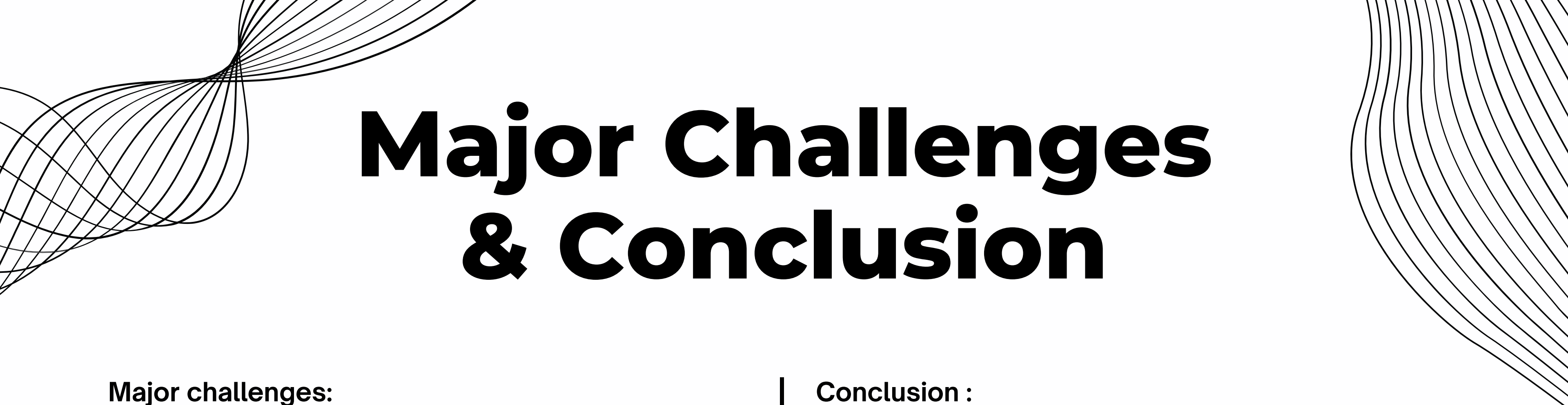
→ keep every layer modular / replaceable / testable =>
Perfect for futur database scaling



The java architecture



A decorative graphic in the top right corner of the page. It features a solid blue square. To its left, a series of thin, black, wavy lines curve downwards and to the left, creating a sense of movement and depth. The lines are closely spaced and follow a similar S-curve pattern.



Major Challenges & Conclusion

Major challenges:

Normalization of the database Hard to deal with all the connections

Manage to connect all the different class : required Organisation + Discipline

Conclusion :

Went from fully mocked memory to a production-ready database

Perfectly ready for scale up

Learned to deal with a test / main environnement in a application